

Project Activity - Regression Analysis

Dataset: California Housing Dataset - `fetch_california_housing` (from `sklearn.datasets`)

Input: 20,640 records, 8 numeric predictors describing California census block groups (median income, house age, average rooms, average bedrooms, population, average occupancy, latitude, longitude)

Target Variable: `MedHouseVal` - median house value for the block group (in \$100,000s)

Objective: Build a complete, end-to-end regression pipeline - from raw data to a validated, tuned model - that predicts median house value from the given census features. Work through every part in order; later parts assume the outputs of earlier ones.

Part A - Dataset Handling

Task A1. Load the dataset and convert it into a pandas DataFrame with properly named columns, including the target column. Display the first 10 rows.

Task A2. Report the shape of the dataset, the data type of each column, and basic descriptive statistics (mean, std, min, max, quartiles) for every numeric column.

Task A3. Check for missing values and duplicate rows. If any are found, handle them appropriately and briefly justify your approach; if none are found, state that explicitly.

Task A4. Identify outliers in at least three features using a method of your choice (e.g. IQR rule or z-score), visualize them with boxplots, and decide - with justification - whether to cap, remove, or retain them.

Part B - Exploratory Data Analysis (EDA)

Task B1. Plot the distribution of the target variable (`MedHouseVal`). Is it symmetric, skewed, or does it show signs of capping (e.g. a spike at the maximum value)?

Task B2. Plot histograms for all numeric features in a single grid of subplots. Note any features that appear heavily skewed.

Task B3. Compute the correlation matrix of all features with the target, and visualize it as a heatmap. Identify the three features most strongly correlated (positively or negatively) with `MedHouseVal`.

Task B4. Create scatter plots of the target against the two or three features you found most correlated in Task B3. Describe the relationship you observe (linear, non-linear, noisy, etc.).

Task B5. Using Latitude and Longitude, create a scatter plot of all block groups colored by `MedHouseVal` (a geographic price map of California). What spatial patterns do you observe?

Task B6. Check for multicollinearity among the predictor features (e.g. using a correlation heatmap between features, or Variance Inflation Factor). Report any pairs of features that are highly correlated with each other.

Part C - Feature Engineering & Preprocessing

Task C1. Engineer at least two new features that could plausibly improve the model, for example: `rooms_per_household` (`AveRooms / AveOccup`-based ratio), `bedrooms_per_room`, or `population_per_household`. Justify why each engineered feature might help.

Task C2. Apply a transformation (e.g. log transform) to any heavily skewed numeric features identified in Task B2, and show a before/after histogram comparison for at least one transformed feature.

Task C3. Split the data into training (80%) and test (20%) sets using `train_test_split` with `random_state=42`. Keep the test set untouched until Part E.

Task C4. Scale all numeric features using `StandardScaler` (or another appropriate scaler), fitting it on the training data only and applying it to both the training and test sets.

Task C5. Assemble your cleaning, feature engineering, and scaling steps into a single scikit-learn Pipeline (or `ColumnTransformer + Pipeline`) so the entire preprocessing flow can be applied consistently and reproducibly.

Part D - Model Development, Training, and Testing

Task D1. Establish a baseline model that always predicts the mean of the training target values. Record its performance (see Part E metrics) as the floor every real model must beat.

Task D2. Train a Linear Regression model on the training data. Report its learned intercept and coefficients, and identify the feature with the largest (absolute) coefficient.

Part E - Model Performance Assessment

Task E1. For model trained in Part D, generate predictions on the held-out test set and compute Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 . Present all results in a single comparison table.

Task E2. For your single best-performing model, create a scatter plot of predicted values vs. actual values, with a diagonal reference line representing perfect prediction.

Task E3. For the same best model, create a residual plot (residuals vs. predicted values). Do the residuals appear randomly scattered around zero, or is there a visible pattern (e.g. funnel shape, curve)? What would that pattern imply about the model's assumptions?

Task E4. Plot a histogram of the residuals for your best model. Do they appear approximately normally distributed?

Part G - Conclusion and Reporting

Task G1. Build one final summary table comparing every model built in this project (baseline, Linear Regression, Ridge, Lasso, Random Forest, Gradient Boosting, and your two tuned models) on MAE, RMSE, and R^2 on the test set.

Task G2. Recommend one model as your final choice for deployment. Justify your recommendation using both predictive performance and practical considerations (interpretability, training time, robustness).

Task G3. Write a short (200–300 word) summary, in plain language suitable for a non-technical stakeholder, explaining what drives median house values in this dataset and how confident they should be in the model's predictions.

Deliverables

Submit a single, well-organized notebook (or script with markdown/comments) containing all code, visualizations, and written answers for Parts A-G, executed top to bottom without errors. Every plot must have a title and labelled axes, and every table should be clearly formatted with column headers.